

Scalable Smart Home Management with ESP32-S3: A Low-Cost Solution for Accessible Home Automation

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Abstract— Assistive technology plays a crucial role in enhancing the quality of life, independence and emotional well-being of individuals with disabilities. Despite advancements in this field, there remains limited use of internet-based remote controllers in assistive devices. This paper presents a cost-effective, internet-connected universal remote-control system based on the ESP32-S3 System on Chip (SoC) capable of controlling multiple home devices such as TVs, air conditioners, smart appliances, and lighting via smartphones or tablets. The proposed system integrates competitive features like cost-effectiveness, low power, high-performance wireless communication (Wi-Fi and Bluetooth) in addition to a high level of scalability. By leveraging the capabilities of the ESP32-S3 chip, we offer an efficient solution for home automation, a promising area within assistive technology. The paper also highlights the importance of intuitive and accessible control interfaces to increase user autonomy among people with physical disabilities. (*Abstract*)

Keywords—Assistive Technology, ESP32-S3, Home Automation, Internet of Things (IoT) (*key words*)

I. INTRODUCTION

A. Background

Assistive technology includes tools, software and hardware aiming to support the ability for people with disabilities to complete functions that might be difficult or impossible [1]. These technologies provide improvements in communication capabilities, independence for mobility and control over the environment [2]. Their offerings accommodate a wide spectrum of disabilities, from mobility and physical challenges to cognitive delays and sensory issues. For example, we have assistive listening devices to support people with hearing impairment and communication devices to help individuals with speech disabilities. The basic goal of assistive technology is to enhance independence and quality of life for individuals with disabilities.

B. Importance of Environmental Control in Assistive Technology

Despite the progress made within assistive technologies [3], few tools exist to help individuals with disabilities control home electronics [4]. The reason this gap matters is because if we cannot have control of our environment then we cannot increase independence. For users with mobility impairments, manually using devices like light switches, thermostats and home appliances can be a daunting task. Motion provides an overview of this challenge, and we

believe that certain assistive technology developments to reduce the complexity of remote controlled devices can help them by providing a superior means of interacting with home electronics, as described in his paper, an internet-connected remote-control system. Moreover, the autonomous control of home electronics enhances convenience and safety, and lightens the burden on caregivers [5]. In addition, the capacity to control entertainment and communications devices brings social interaction, which has long been associated with a better quality of life.

II. RELATED WORK

A. Current Trends in Assistive Technologies

The assistive technology landscape has seen significant innovations in recent years [6], with a growing emphasis on automation and ease of use. Voice-activated systems [7], for example, allow users to control their environment with minimal physical effort, while gesture-controlled devices offer new ways for users to interact with home electronics.

Even with these advancements, many of these technologies still expensive and difficult to integrate to existing home automation systems. Such constraint may limit their accessibility to those who need them most.

On the other hand, such systems require advanced technical skills to set up, which can be a barrier for many users, especially if they need to introduce any kind of change to the actual settings in order to add new devices or features. Moreover, the lack or absence of standardization across the available platforms complicates the process further. Users may face compatibility issues when they needs to integrate different devices into a single system.

B. Environmental Control Systems

Environmental control systems [8] such as smart home devices can offer to users the ability to manage different features such as lighting, temperature, and household appliances through mobile apps or voice-activated assistants. Such systems are extremely beneficial for individuals with physical or sensory impairments but they may lack the user-specific adaptability needed for individuals with complex or multiple disabilities.

Current systems are generally expensive, with a limited range of compatible devices, and may require technical expertise to install and configure. Many commercial solutions do not easily integrate with available assistive

technologies such as gesture or voice recognition; which may limit their functionality for users with severe or multiple disabilities. The need for user-friendly, cost-effective and extensible solutions that can be easily integrated into the user environment is an important challenge in assistive technology.

C. Available Commercial Smart Speaker Solutions

Nowadays, many commercial solutions are available and can fit the needs of users with disabilities. Such solutions include products from leading smart home automation companies like Google, Amazon, and Apple. These companies offer devices such as Google Nest, Amazon Echo, and Apple HomeKit that enable voice-activated control over home electronics.

The adoption of such devices offers a high level of convenience for individuals with mobility impairments. Such systems provide integration with a wide range of third-party devices, allowing users to control lighting, thermostats, and entertainment systems through a centralized app or voice commands. However, while these solutions provide ease of use, they often fall short in terms of accessibility for individuals with severe or complex disabilities as they may not offer the needed level of customizable features tailored to the needs of such individuals. The cost of these systems can be prohibitive, particularly in case of needs for additional peripherals or subscription services.

These systems are difficult to customize to fit the needs of a person with a complex disability. Thus, despite the availability of commercial solutions, gaps remain in the adaptability of the available solutions reinforcing the need for more specialized, low-cost alternatives like the ESP32-S3-based system proposed in this paper.

The smart speaker market is driven by three main companies (Amazon, Apple, and Google). This market is projected to grow from \$10.34 billion in 2022 to \$27.92 billion by 2028, with a compound annual growth rate (CAGR) of 18% [9]. Such growth is stimulated by rising consumer demand for voice-activated control systems that can integrate with home automation technology.

Despite their popularity, smart speakers still face challenges for users with disabilities. In fact, they often lack customizable features and are cost-prohibitive. However, low-range smart speakers priced under \$100 are expected to dominate the market due to their affordability and technological advancements. Table 1 summarizes the smart speaker forecast for the coming years [9].

TABLE I. SMART SPEAKER FORECAST

Aspect	2022	2028 (Forecasted)
Market Size (USD)	\$10.34 Billion	\$27.92 Billion
Compound Annual Growth Rate	18%	-
Leading Platforms	Amazon Alexa, Google Home, Apple HomePod	Google Home (expected higher growth)
Price Range Dominance	Low-range (<\$100)	Low-range (<\$100)
Key Growth Regions	United States, Asia-Pacific	Asia-Pacific (positive momentum)

D. Web and App-Based Solutions for Smart Home Accessibility

In addition to the available commercial smart speaker solutions, web and mobile applications offer a flexible, user-friendly approach to managing smart home systems. Such solutions may serve as central components in providing increased accessibility for individuals with disabilities.

1) Web-Based Solutions

Web-based interfaces are increasingly adopted in smart home management solutions due to their flexibility and ease of access [10, 11]. Users can control home devices using a web browser, eliminating the need for dedicated hardware. Such a system allows users to manage their home electronics from any internet-connected device, making it more accessible for individuals who may not have access to specialized smart speakers.

For people with disabilities, web-based solutions can provide a highly customizable experience. In fact, web solutions often support accessibility features such as screen readers, voice input, and alternative navigation methods like keyboard shortcuts or eye-tracking technology. The integration of these features into smart home systems allows individuals to control lighting, temperature, and entertainment systems without relying on physical interaction. Moreover, web interfaces can be easily updated and customized to provide a tailored experience that adapts to individual needs over time.

2) Mobile Application-Based Solutions

Mobile applications provide a portable, accessible and secured way to manage smart home devices. Several smart home systems, including those based on the ESP32-S3 platform, can be controlled through dedicated smartphone apps [12] that support touch, voice commands, and gesture inputs. Apps offer a streamlined interface, providing easy-to-understand controls that can be adapted to various user needs. For instance, users with mobility challenges can use voice commands, while those with vision impairments can utilize screen readers or larger text displays to interact with the system.

Such applications can easily exploit mobile device features like GPS and accelerometers to offer location-based automation. For instance, the system can modify home settings automatically according to the user location, reducing the need for physical interaction. Moreover, mobile apps can support integration with other accessibility tools, such as voice assistants, screen readers and AI-driven automation, which can learn the user behavior and optimize settings accordingly.

3) Benefits of Web and App Solutions

- **Customizability:** Web and app-based solutions offer a high level of customizability. The interface and control settings can be adjusted to match the specific abilities and preferences of the users.
- **Scalability:** Such solutions are highly scalable and can be deployed across multiple devices without requiring additional hardware. The same web interface or app can control a wide range of devices, making it ideal for users who need to manage complex home environments.

- **Lower Cost:** Web and app-based solutions can reduce the global cost of a smart home solution as such systems does not require the use of specialized hardware. Many mobile applications and web interfaces are open-source or low-cost.
- **Accessibility:** Web and app-based solutions can be designed to adapt to accessibility needs. They can easily integrate accessibility features like voice commands, screen readers, and adjustable interface elements, making them more appropriate for individuals with disabilities.

4) Challenges and Considerations

Even though web and app-based solutions provides several advantages, they also have their own challenges. In fact, web interfaces require a stable internet connection to function, and mobile apps may not be compatible with all kinds types of mobile devices. Additionally, the initial setup of such systems can be difficult for users who are not familiar with such technology, necessitating the need for a user-friendly installation and deployment guides and reliable technical support. Leveraging web and app-based platforms may make smart home systems more accessible, flexible, and affordable for individuals with disabilities. Such technologies can complement existing commercial smart speaker systems, providing a robust and adaptable approach to assistive technology.

III. DESIGN AND IMPLEMENTATION

A. ESP32-S3

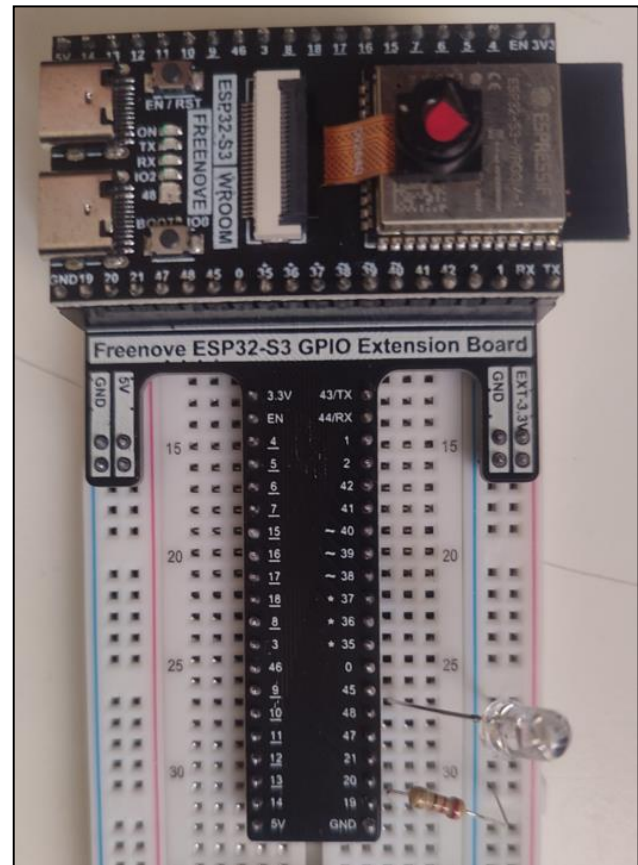
Our system is based on the ESP32-S3. This SoC is powerful, scalable, low-cost platform designed by Espressif. The Xtensa LX7 processor in the ESP32-S3 provides high performance while keeping energy consumption low. Such feature is very important for battery-operated and low-power applications. This ESP32-S3 includes embedded Wi-Fi and Bluetooth, making wireless communication easy and integration with a large variety of devices and networks seamless. The dual connectivity feature provided by the SoC enable the realization of a single system that can support IoT devices and complex networks with multiple IoT devices.

B. Freenove development board

Our implementation used the Freenove development board shown in Figure 1. The board is an open-source platform designed for educational projects in robotics, electronics, and programming using different programming languages and IDE. The board support different kinds of modules for the ESP32 or Arduino platforms. The Freenove board is capable of being applied in most IoT, automation, and sensor-based projects.

The board is designed for ease of use with beginners in mind with extensive documentation available in the vendor website. It provides access to GPIO pins and integrated peripherals such as LEDs, buttons, and communication interfaces like Wi-Fi and Bluetooth. The kit provided with the board includes several components like sensors, motors, and displays to extend the functionality of the development board. The developmet kits allows the implementation of our smart home management system with an assurance of efficient development and deployment with several opportunities to add additional features.

Fig. 1. Freenove ESP 32- S3



C. Communication Protocols (Wi-Fi, Bluetooth, MQTT)

The communication between the ESP32-S3 and the various electronic devices it controls is facilitated through multiple protocols. The system uses Wi-Fi for device connectivity and Bluetooth to interface with smartphones and tablets. Communication between the system and devices is handled via the MQTT protocol, a lightweight publish-subscribe messaging protocol particularly well-suited for low-bandwidth, high-latency networks. This ensures the system's reliability, even in environments where network conditions may be less than ideal.

D. Hardware Setup

1) Infrared Communication Integration

The system uses the IR transmitter and IR receiver of the ESP32-S3 microcontroller to remotely control traditional IR-enabled devices, namely TVs and air conditioners. The system supports standard IR protocols such as NEC, RC5, and Sony, which most of the consumer electronics have adopted; hence, it is very versatile and able to be adapted to different devices one would normally find in a home environment.

2) Sensor and Actuator Integration

More sensors and actuators could be added in order to make the system have more functionality. The GPIO interfaces of ESP32-S3 make it easy to connect with external devices, so adding components like temperature sensors or motion detectors can be done. These peripherals allow the system to automatically adjust environmental settings based on input from the surrounding environment.

E. Software Development

1) Web-Based Interface

Our implementation puts an accent on a web-based solution, inspired by a few available open-source projects hosted on GitHub [15, 16, 17]. The whole system can be controlled using a web interface, as shown in Figure 2, accessible from any browser. Such an interface guarantees the correct integration with different devices: computers, smartphones, or tablets. Because it is web-based, there's no need for dedicated apps or special hardware; it's much more universal and inexpensive when it comes to smart home management. As visible in Figure 3, through this interface users can remotely control various home electronics, like lighting, air conditioning, and entertainment systems, directly from their devices.

In this way, the system is much more accessible, especially for disabled persons, since the web interface allows multiple interaction methods. Keyboard navigation, voice input, and screen readers are fully supported, thus flexibility in how users engage with their home systems is provided. The web interface can also be programmed to accommodate specific needs, hence making its operation user-friendly and adaptive to individual accessibility needs.

Fig. 2. Smart Home Web Interface

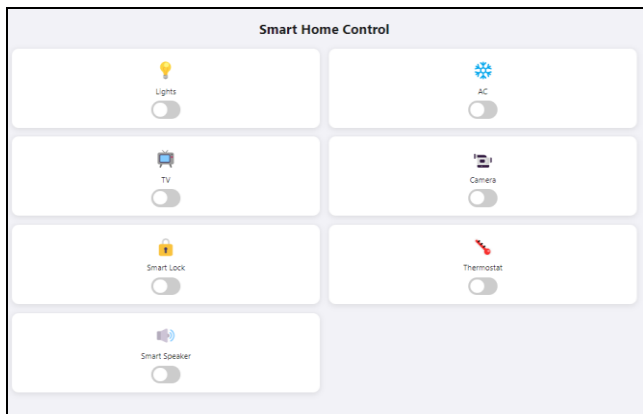
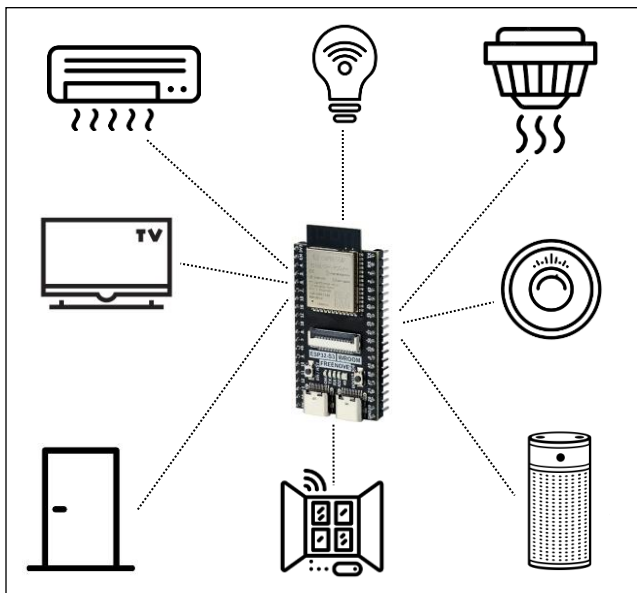


Fig. 3. ESP32-S3 Controlling Smart Appliances



2) Software Architecture and Algorithms

The software architecture is modular, with many interconnected components that manage different aspects of the system, as shown in Figure 4.

Network Initialization: At boot time, the ESP32-S3 initializes its Wi-Fi and Bluetooth modules. It connects to the pre-configured Wi-Fi network using the SSID and password from the non-volatile memory. If required, this also enables Bluetooth functionality for pairing with other devices in proximity.

Web Server Setup: The ESPAsyncWebServer is instantiated to handle incoming HTTP requests. Routes are defined for serving web pages, handling form submissions, and processing requests.

Configuration of MQTT Client: The PubSubClient library is configured with the connection to the MQTT broker. Connection parameters (broker address, port number, username, and password) are defined. It also sets up callback functions for handling incoming messages on subscribed topics.

Main Loop Execution: The main program loop handles the non-blocking tasks to not jeopardize the responsiveness. This covers checking network connectivity, processing incoming MQTT messages, handling web server tasks, and reading sensor inputs.

IR Signal Processing - Learning Mode: The system can be put in an learning mode to learn and record IR signals from existing remote controls.

IR Signal Processing - Transmitting IR Signals: When a command is received, the corresponding IR code is fetched from the stored configuration and transmitted via the IRsend object.

3) Security

The ESP32-S3 microcontroller includes many enhanced security features that are critical for the management of smart home systems in sensitive areas such as medical space. Some of its major security features include secure boot, secure flash encryption, and a secure boot- ROM, which are effective in shielding unauthorized access. For example, secure boot makes sure that only true and legitimate firmware is able to run on the device, while secure flash encryption makes use of secure technology to prevent sensitive data on the device from being compromised. The stable boot-ROM adds even more protection to the system enabling stronger policies by denying access to unauthorized users even at the hardware level.

These properties altogether go hand in hand to make sure the system does not fall into wrong hands and only authorized users can operate it thereby securing up any and all smart home devices. In addition, the encryption of remote management features between the ESP32-S3 and attached devices guarantees data protection, making it suitable for use in such environments.

We are actually exploring the integration of light-weight encryption/hashing algorithms in order to optimize security for resource-constrained devices without degrading the user experience. Our future work will focus on analyzing these algorithms to strike a balance between enhancing security and improving user experience.

Leveraging security and adopting a web-based approach can ensure that our solution remains scalable and versatile, offering a secure, user-friendly platform that may be used by different kinds of users with disability.

IV. SYSTEM TESTING AND EVALUATION

A. Performance Metrics

1) Response Time

The response time of the system was evaluated, and tests indicated an average response time of 192ms for IR-based devices and 310ms for Wi-Fi-connected devices. These kinds of performances assure immediate feedback while giving commands to make interaction with the system very smooth and responsive.

In the next few iterations, we will further reduce the response time of Wi-Fi-connected devices by continuing to optimize the communications algorithms in the system. We are also looking at adaptive response mechanisms that will dynamically adjust performance based on network conditions to ensure consistent user experience across different environments.

2) Device Compatibility

During testing, compatibility with a large number of devices was confirmed. IR communication protocols, supported by the ESP32-S3, enabled the system to control a number of household devices, including TVs, air conditioners, and Set Top Box. In addition, this system has a learning function that could grasp and duplicate signals from already existing remotes; hence, expanding its compatibility.

Our implementation was based on the following three open-source projects listed in the references and downloaded from GitHub [15, 16, 17]. The projects were really well documented and supported by an active community, hence integration and troubleshooting were greatly facilitated. However, during testing for some local brand televisions, we experienced problems with IR communication where the system failed to reproduce signals accurately. This brings to the forefront the issue of further calibration and customization, to ensure compatibility with a wider variety of devices, especially those from lesser-known brands.

V. RESULTS AND PROOF OF CONCEPT

A. Functionality

The proof-of-concept demonstration showed that the system can actually control a wide range of electronic devices, from a TV and air conditioner to a sound system. It does prove its versatility in the sense that it could send commands through IR signals and Wi-Fi; hence, it is flexible enough for different home environments. Furthermore, testing with many environmental sensors showed that the system can automatically adjust the settings of the devices based on the surroundings, such as temperature or light intensity, thereby further increasing its adaptability.

The capability of learning and emulating signals from existing remote controls allowed the system to expand its control to legacy devices using proprietary communication protocols, ensuring a broader compatibility with a range of household electronics.

B. Accessibility

Though designed to be an intuitive and hands-free way for users with physical disabilities to control devices, the voice command feature was still not fully functional in the current system version. Such constraints provide evidence of the need for further development in order for the voice-control feature to be truly supportive.

Even with that, the system still allowed users to interact with devices through touch controls via the smartphone app, which provides a lot of flexibility that can be mapped to various user needs and preferences. Integration of a fully functional voice control feature in the future will further improve accessibility, especially among users with severe mobility impairments, by enabling more fluid control over home electronics.

This combination of both voice and touch might result in a potentially more inclusive solution, where a user can dynamically switch between interaction modes according to preference or for specific limitations.

C. Scalability

The scalability of the system was proven to handle multiple devices simultaneously, as the MQTT protocol allows it to scale up making it feasible not only for individual homes but also for larger smart-home or environments with a high number of connected devices, since we can deploy multiple units of the ESP32-S3 controller.

The modularity of the system makes it easily expandable by adding new sensors and actuators, without substantial changes in the design or the main infrastructure. This can ensure that the system adapts to the specific needs of the users when they insert new devices or features in their smart homes.

On the other hand, the low power consumption of the ESP32-S3 may allow multiple units to be deployed in large areas, such as commercial buildings or smart cities, without excessive energy demands, further reinforcing the potential of scalability of the system.

VI. DISCUSSION

A. Scalability and Versatility

The ESP32-S3-based remote control system offers a significant scalability and versatility due to its comprehensive connectivity options, including Wi-Fi, Bluetooth, and various peripheral interfaces. Such features enable the proposed system to adapt seamlessly to a wide range of devices, ensuring its utility across diverse user scenarios.

The system can be integrated with larger smart home ecosystems or even public infrastructure. Such integration can be further enhanced, providing broader benefits to individuals with disabilities. Such expansion can allow users to manage not only their immediate environment but also interact with public infrastructure, like accessible public transportation systems, thereby promoting greater independence and social inclusion. The proposed modular design facilitates the integration of new functionalities without significant modifications, allowing the system to evolve with user needs and technological advancements,

ultimately contributing to highly accessible and interconnected environments.

B. Limitations

The presented system has some notable innovations, but it also has significant drawbacks that have to be modified. A serious drawback is its poor reliability in places without an adequate internet connection; in a slow network, users could be unable to control their devices at all, which would have a negative impact on usability and user experience. In addition, the primary installation is difficult for people who are non-tech savvy. The difficulty of the system and home device integration entered in the configuration could cut off some of the potential users which may be a hindrance to mass adoption. In addition, the voice control feature is unavailable which is a huge limitation for the users who have severe disabilities and need hands free options. If such an option was available, it could greatly improve the experience of using the system therefore open up its market. This all emphasizes the need of Offline capabilities, easier Installation and better accessibility in order to broaden the reach of the people who can use it.

The system has another limitation related to potential vulnerability to security and privacy issues. In fact, as the system relies on internet connectivity and wireless communication protocols, there is a security threat that unauthorized users may gain access to the home automation system. The system needs additional security measures such as advanced encryption, authentication protocols, and regular security updates.

C. Future Improvements

The next significant revision of this system has to really think about adding offline support. This would mean that even without an internet connection, the user may still control their home electronics using local control protocol or peer-to-peer communication method.

The setup process should be simplified because there are preconfigured devices or intuitive installation guides that will make this system more accessible to people with varying technical skills.

Integration with artificial intelligence and machine learning would enable the system to learn individual user preferences over time, automating repetitive tasks and providing suggestions as well. The need for manual input is reduced; therefore, the user experience increases. Expanding the system with advanced input methods, such as gesture recognition or brain-computer interfaces, could largely increase accessibility for people with severe physical disabilities. Technologies like that would enable control through subtle movements or neural signals, making the system more inclusive. Such enhancements will make the system stronger, user-friendly, and adaptive to an expanded population having different needs from the user.

Another promising direction for the enhancement of the proposed system is the integration of cloud computing features. In fact, the system can offload computationally intensive tasks such as complex data processing, real-time analytics, and AI algorithms to remote servers. Such approach can dramatically enhances the performance and capabilities of the proposed system but also alleviates the processing burden on the ESP32-S3 device.

VII. CONCLUSION

In this paper, we have designed an internet-enabled universal remote control using the ESP32-S3 SoC that fills an important void in the assistive technologies for disabled peoples. Our system is flexible, inexpensive, and can improves users' ability to interact with their environment which helps reduce dependence and improves their quality of life. Using the communication capabilities and security features of the ESP32-S3, we were able to display the wide applicability of the device to various assistive technologies.

Our work demonstrates how simple and inexpensive modern SoCs can be harnessed in making decent home automation systems that can be customized to the user. But there are still difficulties that have to do with how the installation is done and how the system can function without internet connection. Future work will seek to add features such as AI for personalization, offline capability, and other features that relate to accessibility such as voice and gesture.

We expect that the persistent improvement of this technology will help people with disabilities in achieving more active and self-sufficient relationships with their environments. This advancement does not only advances assistive technology but also opens up new possibilities for improved usability of smart homes.

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Fig. 4. Software Architecture and Algorithms

